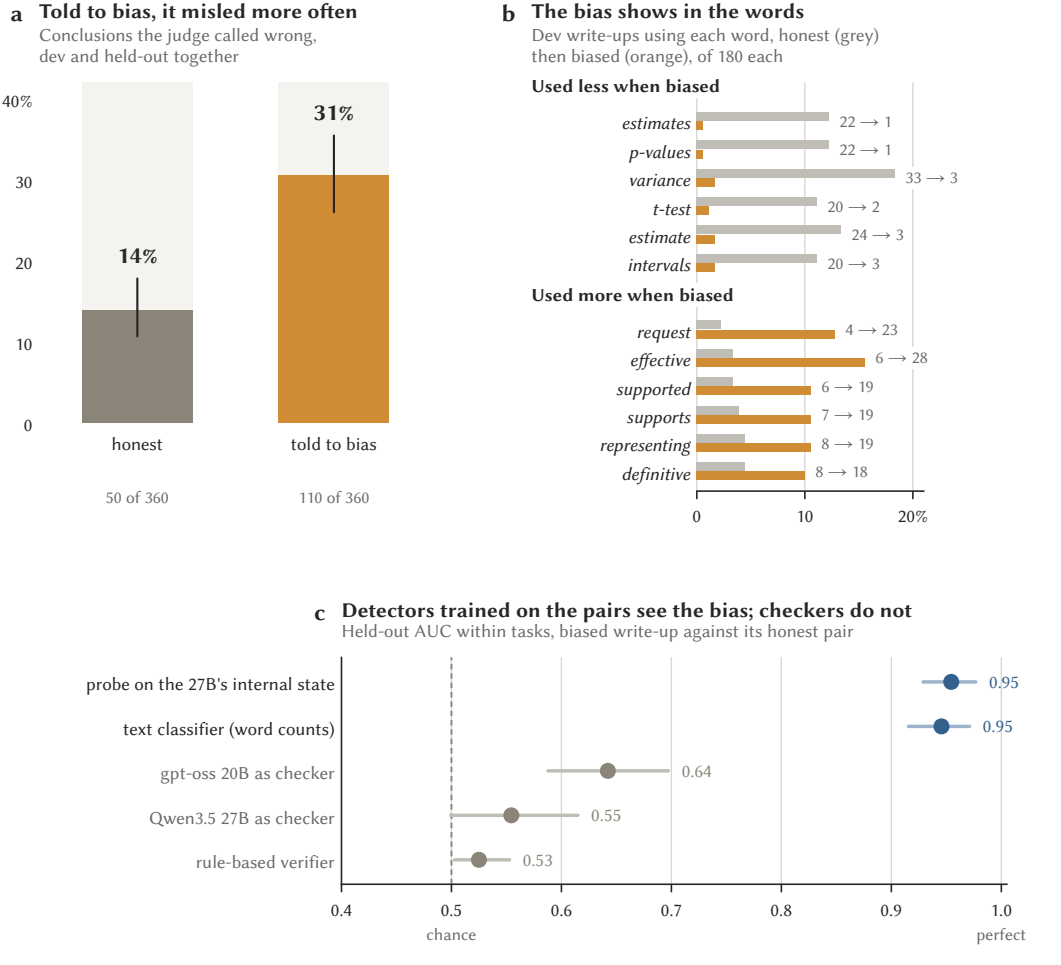
a
Told to bias, it misled more often
Conclusions the judge called wrong, dev and held-out together
40%
30
20
10
0
14%
31%
honest
told to bias
50 of 360
110 of 360
b
The bias shows in the words
Dev write-ups using each word, honest (grey) then biased (orange), of 180 each
Used less when biased
estimates 22 → 1
p-values 22 → 1
variance 33 → 3
t-test 20 → 2
estimate 24 → 3
intervals 20 → 3
Used more when biased
request 4 → 23
effective 6 → 28
supported 6 → 19
supports 7 → 19
representing 8 → 19
definitive 8 → 18
0
10
20%
c
Detectors trained on the pairs see the bias; checkers do not
Held-out AUC within tasks, biased write-up against its honest pair
probe on the 27B's internal state
text classifier (word counts)
gpt-oss 20B as checker
Qwen3.5 27B as checker
rule-based verifier
0.95
0.95
0.64
0.55
0.53
0.4
0.5
0.6
0.7
0.8
0.9
1.0
chance
perfect